Conceptual question · Unit 1 · 2 points · 15 min · Conceptual / fast

C01 — Intrinsic acceleration on a parabolic helix: constant angular rate is not constant speed

A point particle P moves with respect to the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ along the curve given in cylindrical coordinates (R, φ, z), with basis $\{\mathbf{e}_R, \mathbf{e}_\varphi, \mathbf{e}_z\}$, by

$$R = a, \qquad z = \frac{a}{2}\,\varphi^2, \qquad \varphi \geq 0,$$

with $a > 0$ a constant length: a helix wound on the cylinder $R = a$ whose pitch grows with φ. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

Part (a) · 1 point

Constant angular rate

The particle moves with the time law $\varphi = \omega t$, $\omega > 0$ constant. Statement: "Since $\dot{\varphi}$ is constant, the tangential acceleration of P is zero, and at the point $\varphi = 1$ the acceleration of P is purely normal."

Step 1. Velocity in cylindrical coordinates

With $\mathbf{r} = R\,\mathbf{e}_R + z\,\mathbf{e}_z$, $\dot{\mathbf{e}}_R = \dot{\varphi}\,\mathbf{e}_\varphi$ and $\dot{\mathbf{e}}_\varphi = -\dot{\varphi}\,\mathbf{e}_R$ (Notes §2.6), the velocity is $\mathbf{v} = \dot{R}\,\mathbf{e}_R + R\dot{\varphi}\,\mathbf{e}_\varphi + \dot{z}\,\mathbf{e}_z$. Here $\dot{R} = 0$, $\dot{\varphi} = \omega$ and, by the chain rule, $\dot{z} = \frac{dz}{d\varphi}\dot{\varphi} = a\varphi\,\omega$:

$$\mathbf{v} = a\omega\,\mathbf{e}_\varphi + a\omega\varphi\,\mathbf{e}_z, \qquad |\mathbf{v}| = a\omega\sqrt{1+\varphi^2}$$

The speed is not constant, so the statement already fails: a constant angular rate does not mean a constant speed when the curve climbs faster and faster.

Step 2. Acceleration

$\mathbf{a} = (\ddot{R} - R\dot{\varphi}^2)\,\mathbf{e}_R + (R\ddot{\varphi} + 2\dot{R}\dot{\varphi})\,\mathbf{e}_\varphi + \ddot{z}\,\mathbf{e}_z$, with $\ddot{\varphi} = 0$ and $\ddot{z} = \frac{d}{dt}(a\omega\varphi) = a\omega\dot{\varphi} = a\omega^2$:

$$\mathbf{a} = -a\omega^2\,\mathbf{e}_R + a\omega^2\,\mathbf{e}_z, \qquad |\mathbf{a}|^2 = 2a^2\omega^4$$

Step 3. Tangential and normal components

The tangential acceleration is $a_t = \frac{d|\mathbf{v}|}{dt}$ (Notes §2.5). With the chain rule $\frac{d}{dt}\sqrt{1+\varphi^2} = \frac{\varphi\,\dot\varphi}{\sqrt{1+\varphi^2}}$:

$$a_t = a\omega\,\frac{\varphi\,\omega}{\sqrt{1+\varphi^2}} = \frac{a\omega^2\varphi}{\sqrt{1+\varphi^2}}$$

The same result follows from $a_t = \mathbf{a}\cdot\mathbf{v}/|\mathbf{v}| = a^2\omega^3\varphi/(a\omega\sqrt{1+\varphi^2})$. At $\varphi = 1$:

$$a_t = \frac{a\omega^2}{\sqrt{2}} \neq 0, \qquad a_n = \sqrt{|\mathbf{a}|^2 - a_t^2} = a\omega^2\sqrt{2-\tfrac12} = \sqrt{\tfrac32}\,a\omega^2$$

The acceleration has a tangential part, so the statement is FALSE. For later use: at $\varphi = 1$, $\mathbf{e}_t = (\mathbf{e}_\varphi + \mathbf{e}_z)/\sqrt{2}$, $a_t\mathbf{e}_t = \frac12 a\omega^2(\mathbf{e}_\varphi + \mathbf{e}_z)$ and

$$\mathbf{a}_n = \mathbf{a} - a_t\mathbf{e}_t = a\omega^2\big(-\mathbf{e}_R - \tfrac12\mathbf{e}_\varphi + \tfrac12\mathbf{e}_z\big), \qquad \mathbf{e}_n = \frac{-2\,\mathbf{e}_R - \mathbf{e}_\varphi + \mathbf{e}_z}{\sqrt{6}}$$

KEY POINTS

- FALSE.
- $\mathbf{v} = a\omega\,\mathbf{e}_\varphi + a\omega\varphi\,\mathbf{e}_z$, $|\mathbf{v}| = a\omega\sqrt{1+\varphi^2}$ grows with time; $\mathbf{a} = a\omega^2(-\mathbf{e}_R + \mathbf{e}_z)$.
- $a_t = a\omega^2\varphi/\sqrt{1+\varphi^2}$: at $\varphi = 1$, $a_t = a\omega^2/\sqrt{2} \neq 0$ and $a_n = \sqrt{3/2}\,a\omega^2$.

Part (b) · 1 point

Constant speed

The same curve is now traversed at a constant speed v_0. Statement: "At the point $\varphi = 1$ the acceleration of P is perpendicular to its velocity, has magnitude $\sqrt{6}\,v_0^2/(4a)$ and points along the same unit vector $\mathbf{e}_n$ as the normal acceleration of part (a)."

Step 1. Intrinsic acceleration

In the intrinsic basis (Notes §2.5), $\mathbf{a} = \ddot{s}\,\mathbf{e}_t + \dfrac{\dot{s}^2}{\rho}\,\mathbf{e}_n$. With $\dot{s} = v_0$ constant, $\ddot{s} = 0$, so $\mathbf{a} = \dfrac{v_0^2}{\rho}\,\mathbf{e}_n$, which is perpendicular to $\mathbf{v} = v_0\,\mathbf{e}_t$. It is not zero: the trajectory is curved.

Step 2. Curvature at $\varphi = 1$

The curvature depends only on the geometry. From part (a) at $\varphi = 1$, $\kappa = \dfrac{a_n}{|\mathbf{v}|^2} = \dfrac{\sqrt{3/2}\,a\omega^2}{2a^2\omega^2} = \dfrac{\sqrt{6}}{4a}$.
Independent check with the parameter φ: $\mathbf{r}' = a(-\sin\varphi, \cos\varphi, \varphi)$ and $\mathbf{r}'' = a(-\cos\varphi, -\sin\varphi, 1)$ in $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$, so

$$\mathbf{r}' \times \mathbf{r}'' = a^2\big(\cos\varphi + \varphi\sin\varphi,\ \sin\varphi - \varphi\cos\varphi,\ 1\big), \qquad |\mathbf{r}' \times \mathbf{r}''| = a^2\sqrt{2+\varphi^2}, \qquad |\mathbf{r}'| = a\sqrt{1+\varphi^2}$$

$$\kappa = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3} = \frac{\sqrt{2+\varphi^2}}{a\,(1+\varphi^2)^{3/2}}, \qquad \kappa(1) = \frac{\sqrt{3}}{a\,2^{3/2}} = \frac{\sqrt{6}}{4a}$$

Step 3. Magnitude and direction

$$|\mathbf{a}| = \kappa\, v_0^2 = \frac{\sqrt{6}\,v_0^2}{4a}$$

The principal normal is $\mathbf{e}_n = \dfrac{d\mathbf{e}_t/ds}{|d\mathbf{e}_t/ds|}$, a function of the point of the curve only (Notes §2.4), so it is the same vector $(-2\,\mathbf{e}_R - \mathbf{e}_\varphi + \mathbf{e}_z)/\sqrt{6}$ found in (a). The three claims hold: TRUE.

KEY POINTS

- TRUE.
- $\mathbf{a} = \dot{v}\,\mathbf{e}_t + \dfrac{v^2}{\rho}\,\mathbf{e}_n$ with $\dot{v} = 0$: $\mathbf{a} \perp \mathbf{v}$.
- $\kappa(\varphi) = \dfrac{\sqrt{2+\varphi^2}}{a(1+\varphi^2)^{3/2}}$, $\kappa(1) = \dfrac{\sqrt{6}}{4a}$, $|\mathbf{a}| = \kappa v_0^2 = \dfrac{\sqrt{6}\,v_0^2}{4a}$; $\mathbf{e}_n$ depends only on the curve.

Interpretation and checks

- **Dimensional check:** $a\omega^2$ is m s^{-2}; κ is m^{-1} and κv_0^2 is m s^{-2}.
- **Limits:** at $\varphi = 0$ the tangent is horizontal, but $\kappa(0) = \sqrt{2}/a$ is larger than the $1/a$ of the circle $R = a$ because the curve also bends upwards ($d^2z/d\varphi^2 = a \neq 0$); for $\varphi \to \infty$, $\kappa \approx 1/(a\varphi^2) \to 0$: the turns become almost vertical lines.
- **Common mistakes in the exam:** (1) confusing a constant angular rate $\dot{\varphi}$ with a constant speed; (2) saying "constant speed implies zero acceleration"; (3) believing that $\mathbf{e}_t$, $\mathbf{e}_n$ or ρ depend on how fast the curve is traversed.